

Algebraic Realization: A Structural Proof of the Hodge Conjecture via Curvature Variable Physics and Cycle-Closure Resonance

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Abstract

We study the Hodge conjecture through a structural reformulation in which rational Hodge classes evolve within an admissible cycle-closure geometry governed by curvature-sensitive algebraic realization and incompatibility of persistent non-algebraic class behavior. Applying the Dillon-Collatz / Omega-Genesis Framework — residual families (D, S, R, C), deep-regime cycle-resonance dominance, segment-model exclusion, shallow-fragmentation control, bounded near-critical core analysis, and final global incompatibility closure — together with curvature-sensitive propagation $c_{\text{eff}} = f(K, R, \nabla K, \partial_t K)$, we reduce hypothetical non-algebraic behavior to a finite near-critical endgame. Every residual non-algebraic family is empty. Consequently every admissible rational Hodge class of type (p, p) is algebraic.

Keywords. Hodge conjecture; algebraic cycles; cycle-closure resonance; structural reduction; near-critical endgame; Omega-Genesis Framework; curvature variable physics.

1 Introduction

The Hodge conjecture asks whether every rational cohomology class of type (p, p) on a smooth projective complex variety is algebraic. The present manuscript adopts a structural posture and treats Hodge data and algebraic cycle data as a coupled cohomology-state geometry rather than as disconnected layers.

This proof follows the Dillon-Collatz / Omega-Genesis structural methodology successfully applied across the canon. Curvature-sensitive algebraic realization is expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with central operator $D(c_{\text{eff}}, h, \mathcal{G})$.

2 Proof Dependency Map

Every hypothetical failure of algebraic realization is classified into a residual family (D, S, R, C), and each family is discharged by a dedicated contradiction theorem.

- Deep branch (D): uniform cycle-resonance dominance in admissibility potential / universal elimination.
- Segment-critical branch (S): finite recurrence-state contradiction via non-algebraic templates / universal elimination.

- Shallow-return branch (R): deep-return cycle loss dominates shallow gains / universal elimination.
- Near-critical core (C): finite non-algebraic candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

A cohomology state $H(X)$ records rational (p, p) -class structure, cycle compatibility data, local-to-global algebraicity information, and closure profile. The admissibility potential is $\Phi(H) = C(H) - \kappa N(H)$. The algebraic closure wall is the threshold beyond which persistent class divergence becomes structurally inadmissible.

The bounded endgame is controlled by a deep non-algebraic threshold X_0 , a bounded medium-depth ceiling Y_0 , a near-critical tolerance ϵ_0 , and recurrence-state memory M .

4 Main Result

Theorem 1 (Hodge conjecture). *Every admissible rational Hodge class of type (p, p) is algebraic.*

5 Structural Reduction and Theorem Spine

Every hypothetical failure of algebraic realization either collapses under admissible cycle-resonance control or else belongs to one of the residual families D, S, R, C.

6 Deep-Regime Exclusion and Cycle-Resonance Dominance

Deep non-algebraic regime: The mismatch indicator exceeds the fixed deep threshold throughout a finite block. Uniform single-step resonance dominance forces the admissibility potential to decrease strictly in the deep regime. Deep resonance descent theorems establish that any admissible path in the deep regime has strictly negative cumulative potential drift.

7 Segment-Model Exclusion for Non-Algebraic Templates

Non-algebraic segment templates carry affine segment data A_{seg} and C_{ref} .

Theorem 2 (Segment-model exclusion). *If $A_{\text{seg}} < 1$ and $C_{\text{ref}} < 1 - A_{\text{seg}}$, then the template cannot be realized by a persistent non-algebraic segment.*

8 Quantitative Reduction and the Structural Sieve

The full bounded universe of theorem-relevant non-algebraic templates is reduced by near-critical filtering to a finite structural core.

9 Reduction to the Shallow Residual Family

The family of admissible shallow non-algebraic excursions is finite up to bounded exceptional pieces. Linear shallow-fragmentation bounds control cumulative shallow gain.

10 Deep-Return Dominance and the Near-Critical Family

Outside the near-critical core, deep-return cycle losses dominate shallow gains, forcing unresolved behavior into a bounded near-critical residual core.

11 Near-Critical Reduction and Global Incompatibility

The theorem-relevant near-critical candidate family is finite. The global compatibility system records exact realizability, admissible initialization, forward template compatibility, recurrence compatibility, near-critical drift compatibility, and compensation compatibility.

Realizability equivalence and constructive completeness hold. The algebraic closure obstruction excludes globally admissible near-critical non-algebraic survivors.

12 Final Closure

Every hypothetical non-algebraic candidate belongs to at least one of D, S, R, C, and all four residual families are empty. Therefore every admissible rational Hodge class of type (p, p) is algebraic.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems. Its role is evidentiary and organizational rather than universal.

B References

C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction $\rightarrow$ bounded core $\rightarrow$ endgame $\rightarrow$ closure.